MIL-W-43060B

4 August 1971

SUPERSEDING
MIL-W-43060A
12 May 1969

MILITARY SPECIFICATION

WHEELS, ABRASIVE (DEPRESSED CENTER)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type of resinoid bonded, depressed center abrasive wheel.

1.2 Classification.- The abrasive wheel shall be of the following sizes as specified (see 6.2).

Sizes

7 - inch
9 - inch

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 5130

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INCORPORATED

ANSI B7.1 - USA Standard Safety Code for the Use, Care, and Protection of Abrasive Wheels.
ANSI B74.12 - Checking the Size of Abrasive Grain for Grinding Wheels, Polishing and General Industrial Uses.
ANSI B74.13 - Identifying Grinding Wheels and Other Bonded Abrasives, Markings For.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, N.Y. 10018.)

GRINDING WHEEL INSTITUTE

Drawing GWI P.205 - Specification for Mounting, Adapter for 7" and 9" Type 27 Wheels.
Drawing GWI P.216 - Specification for Throw-Away Flange and Nut for Type 27T and 28T.

(Application for copies should be addressed to the Grinding Wheel Institute, 2130 Keith Building, Cleveland, Ohio 44115).

3. REQUIREMENTS

3.1 Standard product.- The abrasive wheels covered by this specification shall be the manufacturer's current, standard, commercial product except for any changes necessary to comply with specification requirements.

3.2 Materials.- Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice provided the completed item complies with all provisions of this specification.

3.3 Construction.-

3.3.1 Abrasives.-

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

3.3.1.1 Abrasive.- The abrasive wheel shall be formed of aluminum oxide in accordance with the manufacturer's current commercial practice (see 6.3).

3.3.1.2 Grain size.- The nominal abrasive grain size and respective symbol shall be in accordance with table I. The size shall be controlled within the limits for sizing of aluminum oxide abrasives as specified in the latest ANSI B74.12 for abrasive grain sizes.

3.3.1.3 Grade and bond.- The wheel grade and bond shall be as specified in table I (see 6.3.1, and 6.3.2). Symbols are in accordance with ANSI B74.13.

3.3.1.4 Reinforcement.- The wheels shall be reinforced with fiberglass cloth, or other reinforcing cloth. Sufficient layers of reinforcement cloth shall be used to meet the ANSI standard B7.1 safety code strength requirements for depressed center reinforced wheels.

3.3.1.5 Thickness.- Thickness shall be in accordance with table I, figure 1, and as specified in 6.2.

Source: https://assist.dla.mil -- Downloaded: 202640-21T20:28Z
Check the source to verify that this is the current version before use.

TABLE I.- Dimensions and symbols

Abrasive 1/	Size (O/A dia.) Inches	Thickness inch	Grain size	Hole Dia. + 0.010 - 0.000	Grade 1/	Bond 1/
Aluminum Oxide (A)	7 + 1/8 - 0	1/8 + 1/16 - 0	24	7/8	U	BF (Resin reinforced)
	7 "	3/16 + 1/16 - 0	24	7/8	Q	"
	7 "	1/4 + 1/16 - 0	24	7/8	Q	"
	7 "	1/4 + 1/16 - 0	24	7/8	U	"
	9 + 1/4 - 1/16	3/16 + 1/16 - 0	24	7/8	Q	"
	9 "	1/4 + 1/16 - 0	24	7/8	Q	"

1/ Symbols are in accordance with ANSI B74.13.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

3.4 Mounting.- The wheels shall be the commercially designated type 27 wheel and shall be capable of being mounted on spindles utilizing adapters specified on Grinding Wheel Institute Drawings GWI P.205, GWI P.216, or equivalents.

3.5 Performance.- The wheels shall comply with speed requirements of ANSI B7.1 with no bond separation, cracking or chipping when tested in accordance with 4.3.1. In order to minimize vibration, the wheels shall comply with the following balance and runout tolerances:

Balance

Diameter	Thickness	Maximum Imbalance
7" and 9"	1/8" and 3/16"	1/8 ounce at the periphery
	1/4"	3/16 ounce at the periphery

Side Runout

Diameter	Thickness	
7"	1/8", 3/16", 1/4"	0.035 TIR Maximum
9"	3/16", 1/4"	0.035 TIR Maximum

3.6 Identification of wheels.- The wheels shall be marked for identification in accordance with ANSI B74.13.

3.7 Workmanship.- The wheels shall be clean with no cracks, chips, gouges,or lumps.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Process inspection.- Inspection shall be made at any point or during any phase of the manufacturing process to determine whether the required operations are accomplished as specified. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specified requirements. Whenever nonconformance is noted, correction shall be made to affected items and process.

a. Grain size is as specified in 3.3.1.2.
b. Grade is as specified in 3.3.1.3.
c. Bond is as specified in 3.3.1.3.
d. Reinforcement is as specified in 3.3.1.4.
e. Wheels are capable of being mounted in accordance with 3.4.

4.2.2 End item inspection.- The lot shall be all the wheels of one size offered for inspection at one time. The sample unit for this examination shall be one complete wheel.

4.2.2.1 Visual examination.- Examination of the wheels shall be made for defects listed in table II. The inspection level shall be II and the Acceptable Quality Level (AQL) shall be 4.0 defects per hundred units.

TABLE II.- Classification of defects

Examine	Defects
Design	Not in accordance with figure 1 or table I. Not reinforced wheel.
Construction	Split, cracked, warped, malformed.
Workmanship	Not clean. Any crack, chip, gouge or lump.
Marking	Incomplete, illegible, or missing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

4.2.2.2 Dimensional examination.- Each sample unit shall be examined for dimensions specified. Any noncompliance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.3 End item testing.- The wheels shall be tested as specified in 4.3.1. The inspection level shall be S-1 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.2.4 Inspection of preparation for delivery.- Preservation, packaging, packing and marking shall be inspected for compliance with paragraph 5.1.

4.3 Tests.-

4.3.1 Performance test.- The abrasive wheel shall be mounted using the proper adapter on a suitable machine and be run at a speed of not less than 5,500 rpm and not exceeding manufacturer's maximum wheel rpm rating, for a period of 2 minutes, to check for imbalance, runout and excessive vibration. Upon completion of this run, the abrasive wheel shall be used to grind a piece of bar stock of wrought iron or carbon steel in accordance with manufacturer's instructions for a period not less than 5 minutes. Bond separation, imbalance, runout, cracking or chipping shall constitute failure of this test (see 3.5).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- Preservation, packaging, packing and marking shall be in accordance with MIL-P-3816. Levels of preservation, packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use.- The abrasive wheel covered by this specification is intended for use on sanders and right-angle-head portable grinders for weld grinding, slotting and snagging.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Thickness required (see 3.4.1.5).
(d) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

6.3 Aluminum oxide.- Aluminum oxide is the medium used in cutting high tensile strength materials such as alloy steel, high speed steel, annealed malleable iron, tough bronze, etc.

6.3.1 Grade is the amount of bond.

6.3.2 Bond is the cohesive medium for holding the grain particles together in the wheel.

Custodians:

Army - GL
Air Force - 84

Review activity:

Army - WC

User activities:

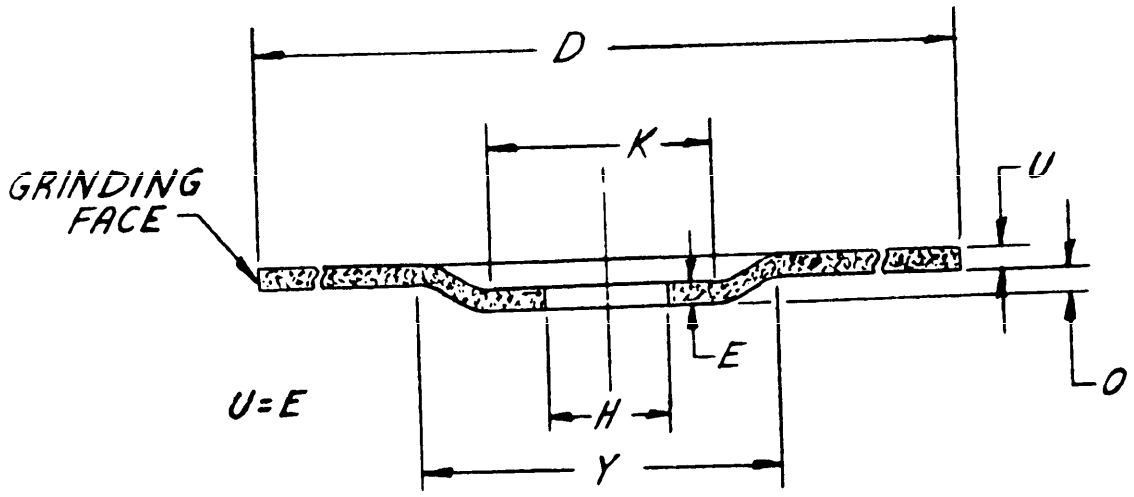
Army - CE
Navy - MC

Preparing activity:

Army - GL

Project No. 5130-0163

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.



D	U & E	H	O	K	Y
7"	1/8"	7/8"	3/16"	1 5/8"	2 5/8"
7"	3/16"	7/8"	3/16"	1 5/8"	2 5/8"
7"	1/4"	7/8"	3/16"	1 5/8"	2 5/8"
7"	1/4"	7/8"	3/16"	1 5/8"	2 5/8"
9"	3/16"	7/8"	3/16"	1 5/8"	2 5/8"
9"	1/4"	7/8"	3/16"	1 5/8"	2 5/8"

FIGURE 1 TYPE 27 WHEEL

Source: https://assist.dla.mil -- Downloaded: 202610-21T20:28Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Wheels, Abrasive (Depressed Center) MIL-W-43060B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:28Z
Check the source to verify that this is the current version before use.